Secure User Authentication Scheme Using Multimodal Biometrics

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Abstract

Authentication is the process of verifying identity. Authentication comprises two steps: Identification and Verification. During identification, users provide information about themselves. In verification, the information the user provides is validated to determine whether the user is legitimate. Over the past decade, the increase in security breaches manifests the security problems associated with authentication involving a single factor, such as password/PIN. Despite being more secure and strongly promoted, two-factor(2FA) or multi-factor (MFA) schemes fail to protect against recent phishing threats such as real-time MITM, control relay MITM, malicious browser-extension-based phishing attacks, and various other attacks and require users to purchase and carry other hardware (security keys or tokens) for additional account protection. Biometrics was adopted in later stages to enhance security and privacy. However, single-modal biometrics have some limitations, and multimodal biometric systems require separate hardware devices to capture individual biometric samples, which incurs extra costs and more time to deploy the solution. In addition, it also affects the user experience, giving rise to less usability. Therefore, a more secure and user-friendly authentication scheme is required. In this paper, an authentication scheme based on multimodal biometrics is proposed. The proposed authentication system uses Face, Voice, and CAPTCHA fetched from a real-time recorded video. This improves the user experience and security compared to other multimodal biometric schemes, where the login credentials are fetched individually.

Keywords— Authentication, Identity, Passwords, Biometrics, 2FA, MFA, Multimodal biometrics

1 Introduction

1.1 Authentication

Authentication is verifying identity to provide access to the system or resources. Digital identity is proof that users provide to the system to prove they are the ones they claim to be. Identity can be either a weak identity or a strong identity. A strong identity is difficult to spoof because identity is the verifier itself. An example is biometrics, in which biometric identity is the verifier. The verifier is considered a strong verifier. There is no separate requirement for a verifier, making it difficult to be spoofed. Whereas in weak identity, there is a requirement for a verifier, making it susceptible to spoofing. An example can be a username, which requires a password as a verifier. Here, the verifier is considered a weak verifier. These credentials are subject to attacks like theft, password guessing, etc. and can be easily compromised. In user authentication, the user proves to the authority that they are the one who they claim to be.

As the world becomes increasingly digitalized, services are more susceptible to cyber-attacks. With more than five billion internet users worldwide[1], authentication is a must to protect their data from cyberattacks. So, there is a vital need for authentication mechanisms which are secure and robust. Authentication is required everywhere. Authentication can be done between (i) User to User, (ii) Device to Device, and (iii) User to Device.

1) **User to User:** In user-to-user authentication, one person provides his identity to the other person, and the other person will act as a verifier to decide whether that user will be given access to the resources.
2) **Device to Device:** In device-to-device authentication, two devices are authenticating each other. This type of authentication is most commonly used in smart vehicles and IoT devices, where devices act as identifiers and verifiers. The rapid development of the Internet of Things (IoT) and edge computing platforms has increased the importance of device-to-device (d2d) authentication. [2] design a D2D continuous authentication protocol that utilizes devices' features (i.e., battery capacity, location, token) to continuously verify devices during the session.
3) **User to Device:** In user-to-device authentication, the user provides their credentials as identity. The. The device will verify and decide whether the user is authorized to access the resources. This is the most commonly used authentication. In this type of authentication, users can use passwords, OTP, fingerprints and a combination of factors to authenticate themselves to access the resources.

The scope of this paper is limited to user-to-device authentication. Authentication can be done either online or offline. According to the World Bank, only about 50% of people in low and middle-income countries have access to the internet (compared to more than 70% of the population in upper-middle-income countries and 89% of people in high-income countries). Globally, Kepios' research suggests that around four in 10 people are not online [3]. In online authentication, users must be connected to the internet to

achieve authentication. This is the conventional method of authentication which is most commonly used.

Authentication factor is used to authenticate individuals. An authentication factor is a distinct security credential employed to authenticate and authorize a user seeking entry, sending messages, or soliciting data from a protected network, system, or application. The most basic authentication mechanism is username and password. There are three authentication factors which are widely used for authentication:
(i) Knowledge-factor
(ii) Possession-factor
(iii) Inherence-factor

1) **Knowledge-factor:** Knowledge-factor refers to those metrics or factors the user knows. These are those factors the user can remember and use the user's knowledge; that is why given name knowledge-factor. Examples of this factor include text passwords and graphical passwords.
2) **Possession factor:** Possession factor refers to those factors which the user owns or presents with the user, such as a smart card or mobile phone. These factors are those factors which can be physically seen and held. These factors have the disadvantage of being stolen.
3) **Inherence factor:** Inherence factor refers to those personal factors inherited by the user and cannot be taken away from the user. These factors uniquely identify each individual and cannot be the same for two persons. Examples include fingerprint, iris, retina, ECG etc.

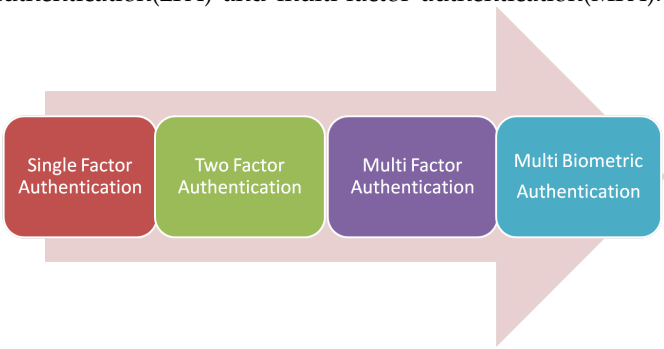
Apart from these three factors, there can be other factors like geographic location, IP address, behaviour factors, etc. But those, as mentioned earlier, three are the most commonly categorized.

Figure 1 shows an authentication scenario using any of the three authentication factors. The steps followed by a user to perform authentication are described as follows:

1) User requests for a particular resource. This resource can be a website, social media account, or digital payment service.
2) This resource access is provided only after submitting specific credentials. These credentials can be either of the three authentication factors discussed above or a combination. So, the website asks for an authentication factor.
3) User submits required authentication factor to the website.
4) Website sends the presented credential to the authentication server for verification.
5) The Authentication server performs the necessary checks to complete the verification. If verification is successful, the website is given a successful verification message.
6) Website then provides the user with access to the requested resource.

1.2 Cataloguing of Authentication schemes

Various authentication schemes have been proposed so far. In the beginning, authentication systems mainly relied on single-factor authentication schemes. The most commonly used single-factor authentication approach is username and password. But passwords are the weakest security mechanisms. The personal data of 815 million Indians exfiltrated from the ICMR's Covid-testing database has been leaked on the dark web in October 2023 [4]. The weakness linked to using passwords as an authentication mechanism leads to the adoption of two-factor authentication (2FA) and multi-factor authentication (MFA) schemes. But people often get confused between 2FA and MFA. Although 2FA and MFA involve using more than one factor for authentication, the two have significant differences. 2FA uses two factors for authentication. For example, the password plus OTP sent through SMS or email is 2FA. At the same time, MFA consists of using two or more independent authentication factors. The two factors should be from different categories. For example, a password plus biometric sample (e.g. Iris) is MFA, but a password in combination with OTP cannot be regarded as MFA. Velasquez, Cora and Rodreguez have found 515 single-factor and 442 multi-factor authentication techniques in their research.[5]
A broad classification of various authentication schemes is shown in figure 2. The schemes of authentication are mainly categorized as Single-factor authentication,two-factor authentication(2FA) and multi-factor authentication(MFA).



1.2.1 *Single factor authentication*

In single-factor authentication, the user uses only one metric for authentication. This metric can either be a knowledge factor (for example, passwords), a possession factor (for example, smart card) or an inherence factor (for example, fingerprint, iris scan, voice, gait, etc.). These are further divided into three types:

1) Knowledge-factor based
   This includes single-factor authentication involving only knowledge factors like passwords, PIN, etc. This is further divided into two categories based on the secret used:
   a) Text-based secret
      This includes those secrets written in text formats like Password and PIN by using alphanumeric characters. The problem with text-based secrets is that users keep the same password for different accounts, which poses a severe security threat such that if one account is compromised, all other accounts are also at risk.
   b) Graphical-based secret
      Graphical-based secrets solve the problem associated with text-based secrets by selecting a set of images

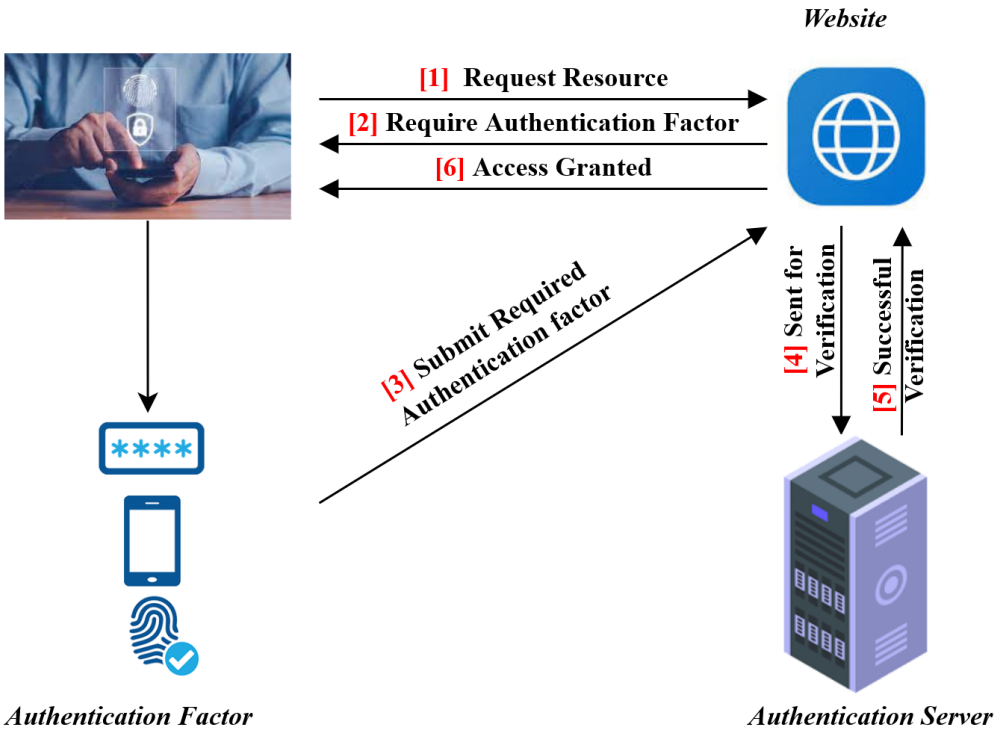


Fig. 1: Authentication Scenario

from the images displayed on the screen and then selecting those images at the time of authentication. It is further categorized into two types:

i) Recognition-based Graphical password
In this authentication scheme, the user must select specific images that they have chosen at registration. One example is Passface [6], in which the user has to determine four face images from a database of faces as their password.

ii) Recall-based Graphical password
In this authentication scheme, the user has to produce the drawing again that they have drawn at the time of registration. An example of this technique is Passpoint [7], where a user can select any location on an image to create a password. A tolerance range is calculated around each selected pixel. To authenticate, the user must click within the tolerance range of their chosen pixels, following the same order.

2) Ownership-factor based
This includes single-factor authentication involving only ownership factors like Smart cards, hardware tokens, mobile phones, OTP, etc.

3) Inheritance-factor based
This includes single-factor authentication involving only inherence factors like fingerprint, face, iris, retina, gait, hand geometry, etc. This is further divided into two categories:

a) Physiological Biometrics
These biometrics include the physical characteristics of an individual. These characteristics are inherent to a person and cannot be changed. Examples are fingerprint, palmprint, Hand geometry, face, iris, retina, ECG, EEG, etc.

b) Behavioral Biometrics
These biometrics include human behaviour and how a person performs a particular action. These are subject to environmental changes. Examples are tapping behaviour, finger gestures, hand gestures, voice, gait, etc.

1.2.2 Two factor authentication

In two-factor authentication, the user uses two metrics for authentication. In addition to one factor, the second factor provides an extra layer of security. The two metrics can be a combination of knowledge and possession factor, knowledge and inherence factor or inherence and possession factor. Two-factor authentication (2FA) enhances security and is designed to protect the user's confidential information from unauthorized users. Passwords are unsafe means of authentication; 2FA requires an additional form of identification to prove you are who you claim to be. The two factors typically used for authentication are something you know (such as a password or PIN) and something you have (such as a mobile device or security token). There is another factor, something you are (Biometrics), which is also included to further enhance the security owing to its advantage over the other two factors. By requiring both factors, 2FA significantly increases the difficulty of someone gaining unauthorized access to your account. There are several reasons why 2FA is required. Some of these are stated below:

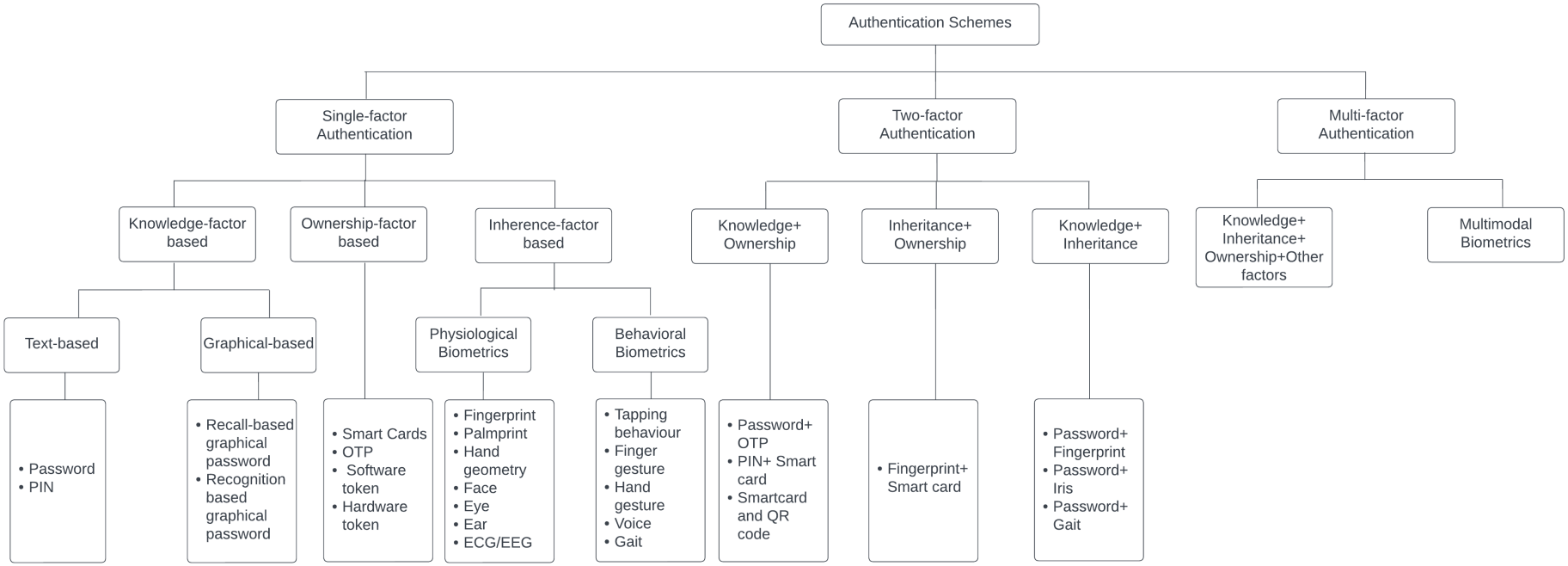


Fig. 2: Cataloguing of Authentication Schemes

1) Passwords can be easily compromised: Passwords are a weak form of authentication because they can be easily guessed, stolen or hacked. 2FA provides extra security to protect the system even if a password has been compromised.
2) Protection against phishing attacks: Phishing attacks are when attackers attempt to trick you into providing your login credentials or other sensitive information. With 2FA, even if an attacker can access the password, they cannot get access until they have a second factor.
3) Compliance: Many regulations and standards require organizations to implement 2FA to protect sensitive information and prevent unauthorized access.

Generally, 2FA is an effective way to meet security needs and protect confidential data. Two factors used in 2FA can be from any of the authentication categories. [8] proposed a 2FA that uses password and OTP. [9] uses smart cards and QR codes as two factors in their authentication scheme. ATMs also employ 2FA by using PINs and smart cards. All these 2FA schemes discussed involve knowledge and ownership factors. Some schemes have also been proposed in the literature that involve a combination of inherence and ownership factors. [10] proposes a scheme that uses fingerprints and smart cards for authentication. There are certain schemes in which knowledge and inheritance factors are combined to verify the identity. Examples include schemes proposed by [11] that use password and fingerprint and by [12] that use PIN and iris.

1.2.3 Multi factor authentication

With the advancement in technology, passwords alone are insufficient to verify identity. Although 2FA adds a second factor to the conventional authentication method, there is no standard for selecting a second factor. The second factor chosen might be attack-prone and can be compromised easily. So, we need to move towards a more secure authentication mode involving two or more authentication factors with the restriction that the factors used should be independent. Multi-factor authentication is shown in figure 3. In Multi-factor authentication (MFA), users need to provide multiple forms of identity to access a system or application. MFA is becoming increasingly important today, where security breaches and identity theft are more common. In multi-factor authentication(MFA), all three factors, viz. knowledge, possession and inherence, can be combined for authentication. Multiple layers of information are added to a single factor, like a password for authentication. The purpose of MFA is to make the authentication process difficult for illegitimate users. MFA allows users to access accounts or systems by providing at least two different forms of identity. This includes:

- What the user knows. e.g. Password or PIN code
- what the user has. e.g. mobile phone or token device
- what the user is, e.g. biometric factors such as fingerprint or facial recognition

MFA is commonly used in online banking, email, social media, and other applications where security is of utmost importance. It is a simple and effective way to enhance the security and protection of any system. For example, the user must submit an OTP and provide their fingerprint scan in addition to the password. Various multi-factor authentication schemes have been proposed in the literature.

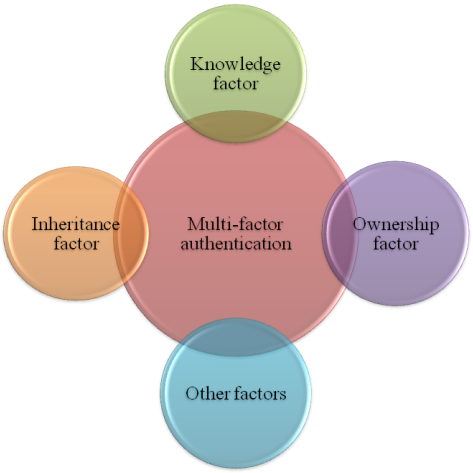


Fig. 3: Multi-factor Authentication

Lee et al. [13] introduced an authentication scheme that leverages smart card and fingerprint verification, improving security and reliability. The fingerprint verification relies on minutiae extraction and matching, while the scheme is based on ElGamal's public key cryptography using two secret keys. Scaria and Mengalingm [14] proposed a multi-factor authentication scheme which comprises three factors - OTP, Fingerprint and noisy password. In this method, a noisy password includes a genuine password concealed within the noise, which only authorized users are aware of. Abdurrahman et al. [15] introduced a multi-factor authentication method called PGT, which is an acronym for pre-shared number, GPS location, and time stamp. This method is designed to boost the security of web accounts by creating a confidential hash code that incorporates the user's current GPS location and a pre-shared number.

1.2.4 Biometric Authentication

Conventional methods of authentication generally involve one or both knowledge and possession factors. However, the problems associated with these factors led to the evolution of biometric authentication. Biometric authentication uses physiological or behavioural traits of human beings to verify identity and authenticate to the system. This type of authentication offers certain advantages over traditional methods. This prevents imposters from attacking the system, as these traits are unique to each individual and cannot be forged. This also provides passwordless authentication, thereby lowering dependence on passwords. However, most systems using biometrics as authentication only employ single biometrics. Using a single biometric factor potentially lowers the authentication system's accuracy rate due to poor data quality, the overlap between identities and limited resources to uniquely identify a person. Unimodal biometrics suffer from the disadvantages mentioned below-

1) Noise: The noise in sensed data can be attributed to defective or poorly maintained sensors, such as a fingerprint sensor that has accumulated dirt, resulting in

deformed and noisy data. Additionally, environmental factors such as cold temperatures can affect voice recognition, while wearing glasses can alter iris recognition performance and variations in lighting conditions can impact facial recognition.

2) Distinctiveness: Another challenge in biometric recognition is distinctiveness, which involves the significant variation of a biometric trait across individuals. Intra-class divergences occur when a user interacts with the sensor incorrectly or exhibits an incorrect facial pose. At the same time, inter-class similarities may be present due to overlaps in the feature sets of multiple users.
3) Non-universality: Non-universality is another issue when the biometric system cannot acquire meaningful data from a particular group of users. For instance, scars or cuts on fingerprints may affect the accuracy of a fingerprint-based biometric system, leading to incorrect minutiae feature extraction. Faulty user-sensor interaction can also result in undesired matching results.
4) Spoofing: Spoof attacks are another challenge wherein an imposter person attempts to spoof the biometric system using fake biometric traits or data from an authorized user. For example, by using artificial fingers or fingerprints.

These problems result in higher false acceptance rates (FAR) and false rejection rates (FRR). Multimodal biometrics has emerged as a solution to some of the issues in unimodal biometrics, offering benefits such as enhanced recognition accuracy, privacy, and improved biometric data enrollment. In this, multiple biometrics features are combined for authentication and are called multimodal biometrics.

1.3 Research problem and motivation

Researchers are trying to find a more secure and robust authentication mechanism to combat security threats and prevent the system from cyber-attacks. Single-factor authentication cannot be a secure authentication method because the password is the main factor used for authentication, and passwords are unsafe and subject to various attacks like guessing and brute-force attacks. Passwords provide the weakest links for the attacker to access the system. MFA and passwordless authentication is gaining popularity and becoming the current trend. In the previous subsection, the benefits of biometric authentication are discussed and how multimodal biometric authentication can be a viable solution. Multimodal biometrics, which involves more than one biometric trait, is a secure and effective authentication mode. All the schemes based on multimodal biometrics involve some problems like

- interacting with different devices to provide biometric identity
- The extra cost involved in purchasing devices

So, there was no scheme which allowed parallel acquisition of biometrics.

1.4 Major Contributions

The proposed scheme provides a secure and higher user acceptability multimodal biometric authentication system, unlike other multimodal biometric-based authentication systems requiring interaction with separate devices or sensors. This scheme involves parallel acquisition of biometric samples, increasing usability, security and deployability. The scheme is entirely passwordless and cost-effective due to no requirement for sensors or separate physical devices to acquire biometric samples. The use of dynamic CAPTCHA reduces video reuse/replay attacks. No prerecorded video can be used because of the availability of this feature. Dynamic CAPTCHA also adds toward the liveliness detection of a specific user because the user will speak the captcha while recording real-time video. This scheme also prevents phishing attacks due to the presence of a captcha because the CAPTCHA is dynamic, and AND rules are implied. So, CAPTCHA validation will fail in case of a phishing attack, thereby preventing unauthorized login. The rest of this paper is organized as follows. The related literature survey is discussed in section 2. The working of the proposed multimodal biometric authentication system is introduced in section 3. Implementation and analysis of the proposed multimodal authentication system are demonstrated in section 4. At last, this paper is concluded by section 5.

2 Related Work

Marasco and Albanese [16] suggested a multi-factor authentication approach called FingerPIN, which combines knowledge and inherence factors with a fingerprint-based personal identification number(PIN). The improved version of the FingerPIN scheme [17] strengthens the security of this scheme and makes it more user-friendly. Li and Hwang [18] proposed a remote user authentication scheme based on biometrics which uses smart cards and relies on biometric verification, a one-way hash function, and smart card technology. Das [19] improved the work of Li et al. and proposed a new scheme in 2011. Das's scheme does not provide resistance against the offline password guessing attack or the privileged insider attack and does not provide user anonymity. To address the limitations in Das's scheme, Lee and Hsu [20] introduced an alternative approach that employs extended chaotic maps.

D. Jagadiswary and D. Saraswady [21] proposed a scheme that combines the RSA asymmetric cryptographic algorithm with biometric traits like fingerprints, retina scans, and finger vein patterns. Puneet Gupta et al. [6] devised a precise authentication system that utilizes finger vein technology to elevate the quality standards of vein images and mitigate noise data during extraction. Das et al. introduced a finger-vein-based Biometric Identification System that uses a Convolutional Neural Network (CNN) [8]. Still, it does not provide robust security for complex databases, particularly in cross-matching scenarios.

M. Hammad et al. [22] propose two multimodal authentication systems (sequential and parallel) and use a combination of ECG and Fingerprint. They apply Convolution Neural Network (CNN) and Q-Gaussian multi-support vector machine (QG-MSVM). Using two datasets, they attained an EER of 0.14% and 0.10% for the sequential systems. The parallel system achieves an EER of 0.40% and 0.32% for respective datasets.

Kim and Hong [23] propose a multimodal biometric authentication approach using teeth image and voice as biometric features. The individual matching scores obtained

from the teeth image and voice are combined using a weighted summation operation, and the fused score is used to decide the acceptance or rejection. The proposed method produces results with an EER of 2.13%.

Ahamed et al. [24] propose an AI-based multimodal biometric authentication model for device-level authentication that involves ECG (Electrocardiogram) and PPG (Photoplethysmography) signals. This authentication model provides results with 99.8% accuracy and an Equal Error Rate (EER) of 0.16.

[25] propose a scheme that combines fingerprint, face, and iris biometrics based on matching scores. The SVM classifier, the sparse SVM classifier, and the rough set fuzzy classifier are used for fusion and classification purposes.

Multimodal schemes proposed so far offer less usability experience and could be more cost-effective. Because user needs to submit multiple biometric samples separately, and their acquisition requires costly hardware devices or sensors. So, we design an authentication scheme that offers parallel acquisition of biometric samples and does not require expensive hardware devices or sensors. Our scheme also provides resistance to phishing attacks.

3 The Proposed Scheme

The proposed scheme relates to multimodal biometric systems. Notably, it specifies a more secure and user-friendly biometric authentication system. This scheme provides a multimodal biometric authentication system that combines face and voice. In addition to biometrics, a dynamic CAPTCHA was also present. Each biometric feature proposed in the scheme and the CAPTCHA was extracted from the real-time recorded video. The three credentials used in the scheme are matched according to the AND logic rules. This ensures that successful authentication is achieved only when all the credentials are compared positively. The present invention also provides an authentication system that involves the parallel acquisition of biometric features.
The proposed authentication scheme's registration phase considers Email address and unique username. After this, the user records the real-time video while speaking the dynamic CAPTCHA. This recorded video is saved in the database. The proposed authentication system uses biometric features of face and voice. The user is required to capture a real-time video. However, while recording the video, the user must speak the dynamic CAPTCHA displayed on the screen. From this real-time recorded video, face, voice and CAPTCHA are extracted. Similarly, face and voice are extracted from real-time video recorded at the time of registration. These face and voice features extracted are then subjected to matching with the features extracted from registration time real-time recorded video corresponding to a unique username using an appropriate matching algorithm of choice. Also, the voice/audio is converted to text and matched with the dynamic CAPTCHA text displayed on the screen. All these matching results are then applied AND logic, which ensures positive authentication only if all the results are positive. These results are combined to decide whether the authentication is successful and give the final decision. Dynamic CAPTCHA reduces video reuse/replay attacks and adds to the liveliness detection of a specific user.

Initially, we start with entering a unique username that helps in user identification and then a real-time video capturing field and dynamic CAPTCHA is displayed. The individual trying to authenticate themself records real-time video while speaking dynamic CAPTCHA, displayed on the screen. From this recorded video, frames are extracted. Also, audio is extracted from this recorded video. Extracted frames are subjected to matching with the frames of the video submitted at the time of registration corresponding to the unique username submitted. Similarly, audio extracted from the login video and the audio extracted from the video stored in the database corresponding to the unique username submitted at the time of registration for the user is matched to know if the user is the one who is speaking in the video. Also, the extracted audio is converted to text matched with the CAPTCHA displayed for CAPTCHA validation checking. After matching frames and audio is complete, the matching score is checked to see if it is above a certain threshold and matching is successful. Then, all these results are subjected to AND logic, which gives positive results only if all validations and matching are positive. The user is granted access to resources if the result is positive. Otherwise, access is not granted.

There are three phases involved in this authentication scheme -

- Registration Phase
- Login Phase
- Authentication phase

3.1 Registration Phase

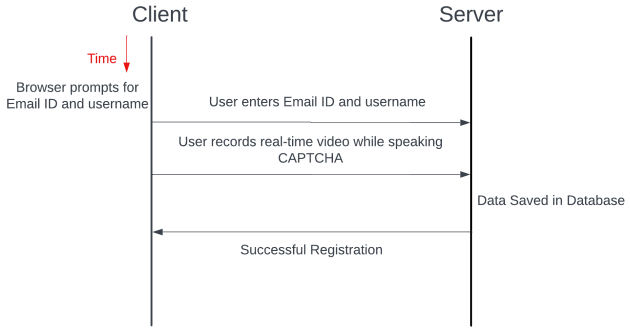


Fig. 4: Registration Phase

Following steps are involved in the registration phase:

1) Initially, the user enters their personal information - Email ID username. The username and Email ID needs to be unique for each individual.
2) In the next step, a dynamic CAPTCHA is displayed on the screen, and a video recording interface is presented. The user has to record their video while speaking the CAPTCHA showed.
3) On clicking the start recording button, the video recording will start. On clicking stop recording, the recorded video is available for preview.
4) On clicking save, this video link gets saved in the database for its requirement during authentication, and

the video is saved on the server side in the format "username.mp4" in a folder

5) On clicking the register button, the user is registered successfully if the submitted credentials pass all the checks implied. Users can further update their information like address, full name, etc.

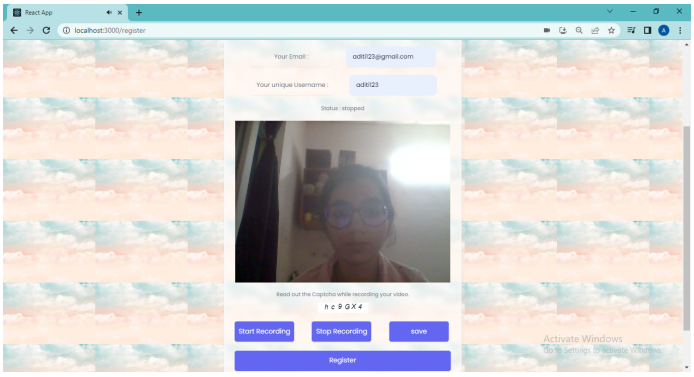


Fig. 5: Registration page after entering credentials

3.2 Login Phase

The following steps are involved in the login phase:

1) The user enters the username submitted at registration. If the username does not match that submitted at registration, we cannot proceed further.
2) A dynamic CAPTCHA is also displayed on the screen. This is server-side CAPTCHA, which is dynamic.
3) The User trying to log in needs to record a video while speaking the CAPTCHA displayed.
4) On clicking the start recording button, the video recording will start. On clicking stop recording, the recorded video is available for preview.
5) On clicking save, the login video is saved on the server side in the format "username_l.mp4" in a folder different from the one in which registration videos are stored.

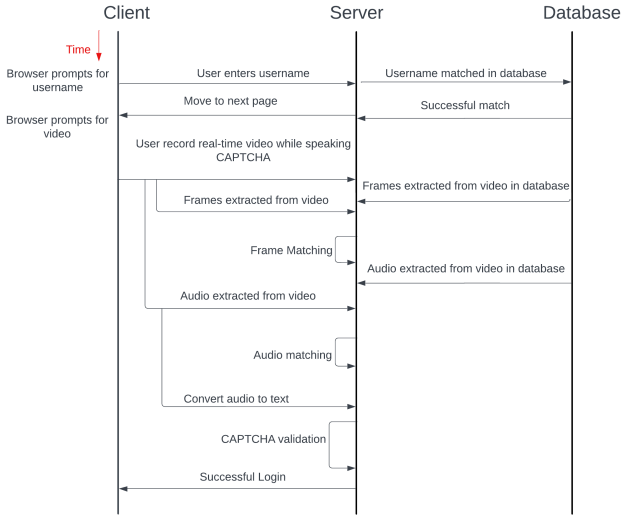


Fig. 6: Login and Authentication Phase

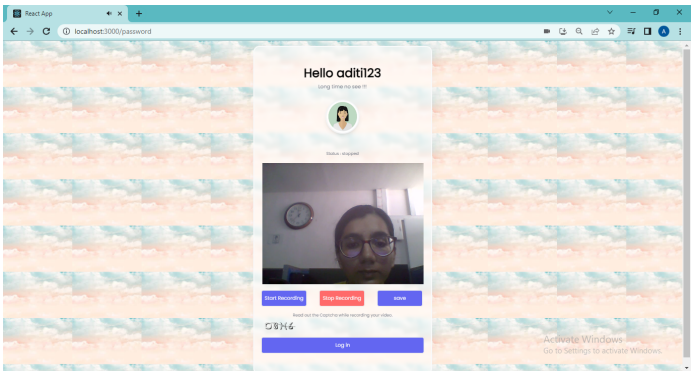


Fig. 7: Login Steps

3.3 Authentication Phase

The following steps are involved in the authentication phase:

1) Initially, the username is matched whether it matches with that submitted in registration.
2) After the video is recorded and the login button is clicked, processing will take place on the server side.
3) Corresponding to username, video path is fetched from the database, and video is fetched at that path.
4) Login video is fetched from its path.
5) Frames are extracted from the video retrieved in step 3.
6) Similarly, frames are extracted from the login video obtained in step 4.
7) Frames obtained in steps 5 and step 6 are subject to matching using a matching algorithm.
8) If step 7 result is positive, that is, frames are matched, then audio is extracted from video obtained in step 3 and stored as "username.mp3".
9) Similarly, audio is extracted from step 4 video and stored as "username_l.mp3".
10) Audios obtained in step 8 and step 9 are then subjected to matching using audio matching algorithm.
11) If the step 10 result is successful, then the audio obtained in step 9 is converted to text and subject to CAPTCHA validation check.
12) If the step 11 result is positive, then login is successful.

4 Implementation and Analysis

4.1 Experimental Setup

To perform authentication, a website was developed which shows the authentication steps involved for checking the real-time working of the proposed scheme.

4.1.1 Hardware Requirements

The Laptop used for experimenting was an Intel(R) Core(TM) i5-4210U CPU @ 1.70GHz 2.40 GHz with 8.00 GB installed RAM. The system was a 64-bit operating system, x64-based processor. The system was running Windows Operating system with edition Windows 10 Education, version 22H2, and OS build 19045.2965. The website was hosted on localhost in Chrome browser Version 114.0.5735.110 (Official Build) (64-bit).

4.1.2 Software Requirements

The experiment requires the editor Visual Studio Code Version 1.79.0 (user setup). The front end of the proposed scheme was built in React JS, and the back end was built in Node JS. This experiment requires Node.js 16.17.1. The database used is MongoDB, and we use MongoDB Atlas. The experiment further requires some packages like FFMPEG [26], speechRecognition [27] library, librosa [28] for audio-related tasks and svgCaptcha [29] for server-side CAPTCHA generation.

4.2 Analysis of the proposed scheme

We have used the Bonneau et al. framework [11] to analyse our proposed scheme. This comparison is done to access our scheme in terms of usability, deployability and security in comparison to existing authentication schemes. We have included Uname-Passwd in our comparison as it is the most easy form of authentication and preferred by normal users. Hardware token based schemes like Yubikey U2F and FIDO2 are being compared because they offer more security although they have overhead of being carried everytime and costly. Xie et al.'s CamAuth, a two factor authentication scheme which prevents various attacks Like MITM and phishing attacks and considered secure web authentication scheme is also considered for comparison. [15] is a MFA scheme included to compare our scheme's benefits against MFA schemes. Next we compare our scheme against other multimodal biometric authentication schemes like [10], [22], [23] already existing in the literature. This is described in the table 1 given below:

5 Conclusions

This paper proposes a multimodal biometric-based authentication system that enhances usability by allowing parallel acquisition of biometric samples. This scheme relates to a user authentication system based on a dynamic CAPTCHA and a real-time single Video input to develop a multimodal biometric authentication system. Through our research, we have achieved the following key findings:

1) We have found that no proposed scheme involves the parallel acquisition of biometric samples and checking liveliness. We have designed a scheme which provides the same.
2) The schemes already present in the literature in the field of multimodal biometrics are costly due to the use of expensive devices, which hinders their applicability by users. Our scheme is cost-effective.

These findings provide valuable insights into multimodal biometric authentication schemes and contribute to the existing body of knowledge in several ways. They offer a deeper understanding of the problem and lay the foundation for further advancements and improvements in the future.
Moreover, our research has highlighted the significance of multimodal biometrics and its impact on authentication. By addressing and investigating this aspect, we have made significant progress towards advancing the field and filling the existing research gaps.

Project_1 > login_system > backend > extractFrames.py > ...

```python
file =sys.argv[1]
# Read the video from specified path
cam = cv2.VideoCapture(file)

# creating a folder named data
try:
    if not os.path.exists('data'):
        os.makedirs('data')
except OSError:
    print ('Error: Creating directory of data')

# Reading frames
currentframe = 0
while(True):
    ret,frame = cam.read()
    if ret:
        # if video is still left continue creating images
        name = './data/frame' + str(currentframe) + '.jpg'
        # writing the extracted images
        cv2.imwrite(name, frame)
        currentframe += 1
    else:
        break
# Release all space and windows once done
cam.release()
cv2.destroyAllWindows()
```

Fig. 8: Frame Extraction

roject_1 > login_system > backend > extractAudio.py > convert_video_to_audio_ffmpeg

```python
import subprocess
import os
import sys

video_file=sys.argv[1]

#Converts video to audio directly using `ffmpeg` command with the help of subprocess module
def convert_video_to_audio_ffmpeg(video_file, output_ext="mp3"):
    filename, ext = os.path.splitext(video_file)
    subprocess.call(["ffmpeg", "-y", "-i", video_file, f"{filename}.{output_ext}"],
                    stdout=subprocess.DEVNULL,
                    stderr=subprocess.STDOUT)

if __name__ == "__main__":
    vf = sys.argv[1]
    convert_video_to_audio_ffmpeg(vf)
```

Fig. 9: Audio Extraction

Project_1 > login_system > backend > matchAudio.py > ...

```python
import librosa
import numpy as np
from sklearn.metrics import pairwise_distances
import sys

file1=sys.argv[1]
file2 =sys.argv[2]
audio_file_1, sr_1 = librosa.load(r"{}".format(file1))
audio_file_2, sr_2 = librosa.load(r"{}".format(file2))

mfcc_1 = librosa.feature.mfcc(y=audio_file_1, sr=sr_1, n_mfcc=13)
mfcc_2 = librosa.feature.mfcc(y=audio_file_2, sr=sr_2, n_mfcc=13)

distance = pairwise_distances(mfcc_1.T, mfcc_2.T, metric='euclidean')

threshold = 200
if (distance < threshold).all:
    # print("The audio files are from the same person.")
    print("yes")
else:
    # print("The audio files are from different persons.")
    print("no")
```

Fig. 10: Audio Matching

Aspect	Benefits offered	Uname-Passwd	Xie et al.	Abdurrah-man et al.	Sivakumar et al.	Hamad et al.	Kim and Hong	Yubikey U2F	FIDO2	Proposed scheme
Usability	Memorywise-effortless	✗	✗	❑	❑	✓	✓	✗	✓	✓
	Scalable-for-Users	✓	✗	❑	✗	✓	✓	❑	✓	❑
	Nothing-to-Carry	✓	❑	✓	✓	✓	✓	✗	✗	✓
	Physically-Effortless	✓	✗	✓	✗	✗	✗	❑	✓	❑
	Easy-to-Learn	✓	❑	✓	✓	✓	✓	✓	✓	✓
	Efficient-to-Use	✓	❑	✓	❑	✗	❑	✓	✓	❑
	Infrequent-Errors	✓	✗	✓	✓	❑	✓	✓	✓	✓
	Easy-Recovery-from-Loss	✓	❑	✓	✓	✗	✗	❑	✗	✓
Deployability	Accessible	✓	✗	✓	✗	❑	❑	✓	✓	❑
	Negligible-Cost-per-User	✓	❑	❑	✗	✗	✓	✗	❑	✓
	Server-Compatible	✓	✗	❑	✓	✓	✓	✗	✗	✓
	Browser-Compatible	✓	❑	✓	✓	✓	✓	❑	✓	✓
	Mature	✓	✗	✗	✗	❑	✗	✓	✓	✗
	Non-proprietary	✓	✓	✓	✓	✓	✓	✗	✓	✓
Security	Resilient-to-Physical-Observation	✗	✓	✓	✓	✓	✓	✓	✓	✓
	Resilient-to-Targeted-Impersonation	✗	✓		✓	✓	❑	✓	✓	✓
	Resilient-to-Throttled-Guessing	✗	✓	✓	✓	✓	✓	✓	✓	✓
	Resilient-to-Unthrottled-Guessing	✗	✓	✓	✓	✓	✓	✓	✓	✓
	Resilient-to-Internal-Observation	✗	✗	✓	✓	✓	✓	❑	✓	✓
	Resilient-to-Leaks-from-Other-Verifiers	✓	✓	✓	✗	✓	✓	❑	✓	✓
	Resilient-to-Phishing	✗	✗	✗	✓	❑	❑	✓	✓	✓
	Resilient-to-Theft	✓	✓	✓	✓	✓	✓	✓	✗	✓
	No-trusted-Third-Party	✓	✓	✓	✓	✓	✓	✗	✓	✓
	Require-Explicit-Consent	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Unlinkable	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Benefit offered count	***18***	***10***	***18***	***17***	***17***	***18***	***13***	***20***	***20***

TABLE 1: Analysis of the Proposed Scheme where ✓: Benefit offered by the scheme, ❑: Partial benefit offered by the scheme ✗: Benefit not offered by the scheme

```python
def hamming_distance(phash_dict1, phash_dict2):
    total_distance = 0
    num_pairs = 0

    for key1 in phash_dict1:
        if key1 in phash_dict2:
            binary_str1 = bin(phash_dict1[key1])[2:].zfill(64)
            binary_str2 = bin(phash_dict2[key1])[2:].zfill(64)

            distance = sum(1 for bit1, bit2 in zip(binary_str1, binary_str2) if bit1 != bit2)
            total_distance += distance
            num_pairs += 1

    return total_distance / num_pairs
```

Fig. 11: Hamming Distance used in Frame Matching

```python
directory_path = "./data"
phash_dict1 = {}
for filename in os.listdir(directory_path):
    if filename.endswith(".png") or filename.endswith(".jpg"):
        file_path = os.path.join(directory_path, filename)
        img = Image.open(file_path)
        phash = str(imagehash.phash(img))
        phash =int(phash,16)
        phash_dict1[filename] = phash

directory_path1 = "./datal"
phash_dict2 = {}
for filename in os.listdir(directory_path1):
    if filename.endswith(".png") or filename.endswith(".jpg"):
        file_path = os.path.join(directory_path1, filename)
        img = Image.open(file_path)
        phash = str(imagehash.phash(img))
        phash =int(phash,16)
        phash_dict2[filename] = phash

# Calculate hamming distance between phash dictionaries
distance = hamming_distance(phash_dict1, phash_dict2)

threshold = 25
if distance <= threshold:
    # print("The phash dictionaries match.")
    print("yes")
else:
    # print("The phash dictionaries do not match.")
    print("no")

shutil.rmtree("./data")
```

Fig. 12: Frame Matching

```python
import speech_recognition as sr
from os import path
from pydub import AudioSegment
import sys
import time
file=sys.argv[1]

# convert mp3 file to wav
sound = AudioSegment.from_mp3(file)
sound.export("transcript.wav", format="wav")

# transcribe audio file
AUDIO_FILE = "transcript.wav"

# use the audio file as the audio source
r = sr.Recognizer()
with sr.AudioFile(AUDIO_FILE) as source:
    audio = r.record(source)  # read the entire audio file
    print("Captcha: " + r.recognize_google(audio))
```

Fig. 13: CAPTCHA(Audio) to text

TABLE 2: NOTATIONS AND ABBREVIATIONS

Notation	Description
$Uname$	Username
V_r	Video captured during registration phase
V_l	Video captured during login phase
F_r	Frames extracted from registration video
F_l	Frames extracted from login video
FM_result	Frame matching algorithm result
A_r	Audio extracted from registration video
A_r	Audio extracted from login video
AM_result	Audio matching algorithm result
P_1, P_2	Perceptual Hash value
$MFCC$	Mel-frequency Cepstral Coefficients
$pHash$	Perceptual Hashing Algorithm for images

Algorithm 1 Login And Authentication Procedure

Input: V_r : Video captured during registration phase
Output: Authentication Successful message

$Uname$: Initially, user enters username and it is authenticated
V_l = Captured during login phase
F_r = Frame extraction (V_r) using OpenCV's $cv2.VideoCapture$ class
F_l = Frame extraction (V_l) using OpenCV's $cv2.VideoCapture$ class
FM_result = Frame Matching (F_r, F_l)
if $FM_result\ is\ true$ **then**:
    A_r = Audio Extraction (V_r) using $ffmpeg$ framework
    A_l = Audio Extraction (V_l) using $ffmpeg$ framework
    AM_result = Audio Matching (A_r, A_l)
end if
if $AM_result\ is\ true$ **then**:
    if $CAPTCHA_Validation\ is\ successful$ **then**:
        $LoginSuccessful$
    end if
end if

Algorithm 2 Frame Matching

Input: F_r, F_l
Output: Frame Matching Result (Yes or No)

for each frame in F_r **do**
    $P_1 = Calculate\ PerceptualHash$
    $Dict_1 = P_1$
end for
for each frame in F_l **do**
    $P_2 = Calculate\ PerceptualHash$
    $Dict_2 = P_2$
end for
$hd = hamming_distance(Dict_1, Dict_2)$
if $hd < threshold$ **then**
    $Frames_match$
else
    $Frames_not_match$
end if

Algorithm 3 Audio Matching

Input: A_r : Audio extracted from registration video
A_l : Audio extracted from login audio
Output: Audio Matching Result (Yes or No)
AF_r = Calculate audio waveform from A_r using librosa library
AF_l = Calculate audio waveform from A_l using librosa library
$MFCC_r = calculate_mfcc(AF_r)$
$MFCC_l = calculate_mfcc(AF_l)$
$ED = euclidean_distance(MFCC_r, MFCC_l)$
if $ED < threshold$ **then**
    $Audio_match$
else
    $Audio_not_match$
end if

5.1 Frame Extraction and Frame Matching

OpenCV, short for Open Source Computer Vision Library, is a powerful and widely used library for real-time computer vision applications. We are using OpenCV's $cv2.VideoCapture$ class available in python to extract frames from videos.

We are calculating $pHash$ of each frame captured from video during frame extraction step. A perceptual hash is a compact digital fingerprint derived from a multimedia file (image, audio, video). Unlike cryptographic hashes (like MD5 or SHA256), it focuses on capturing the perceptual similarity of the content, not the exact data. This means that even if an image is slightly modified (e.g., resized, compressed, brightness adjusted), the perceptual hash remains similar, as long as the overall visual content is preserved. Perceptual hashing algorithms typically involve these steps:

- Preprocessing:The multimedia file might undergo basic transformations (e.g., resizing, color space conversion) to ensure consistency.
- Feature Extraction: The algorithm extracts features that capture the essence of the content. For images, these features include Edges, colors, spatial patterns, textures.
- Hashing Function: The extracted features are fed into a hashing function that generates a compact digest (the perceptual hash). This function is designed to produce similar hashes for perceptually similar content.

$pHash$ is the perceptual hashing algorithm popularly used for images. We calculate hamming distance between the phash calculated by first converting these phashes to binary strings to decide the frame matching result. Hamming Distance compares two binary strings of equal length (like phash outputs). It calculates the number of positions where the corresponding bits differ (mismatches). Lower Hamming distance indicates higher similarity between the phashes.

5.2 Audio Extraction and Audio Matching

We are using FFMPEG frame work to extract frames from video captured during registration and login phase. FFmpeg provides a powerful tool for extracting audio from videos. We can use it to quickly grab the audio stream without re-encoding. Alternatively, ffmpeg also allows re-encoding the audio into a different format like MP3. We are using mp3 format and store audio file as $username.mp3$.

For audio matching, we are using Librosa library available in python. It is a versatile tool for working with audio data in various analysis and processing tasks. We use $librosa.load$ to load audio files. It Loads an audio file from disk (specified audio path) into a NumPy array representing the audio waveform. We then compute Mel-frequency cepstral coefficients (MFCCs) from an audio signal. MFCCs are a popular feature representation in audio analysis, capturing the spectral envelope of the sound. One of the parameters in MFCC calculation is n_mfcc which represents the number of MFCC coefficients to compute. We have set this parameter value to 13. This determines the dimensionality of the output feature vector.MFCCs are widely used in various audio processing tasks, including - Speech recognition, Speaker identification. We compute the Euclidean distance between two MFCC vectors to measure their similarity. Smaller distances indicate a higher degree of similarity in the spectral characteristics of the corresponding audio segments.

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